Current Directions in Psychological Science
21(5) 290–296
© The Author(s) 2012
Reprints and permission:
sagepub.com/journalsPermissions.nav
DOI: 10.1177/0963721412452557
http://cdps.sagepub.com

Nature and Nurture in Personality Development: The Case of Neuroticism and Extraversion

Christian Kandler
Bielefeld University

Abstract

The stability of interindividual differences (i.e., rank-order continuity) in personality traits tends to increase with age until it plateaus in middle adulthood and finally decreases in old age. Rank-order continuity also tends to decrease as the time intervals between occasions of personality assessment increase, irrespective of age. These patterns show that personality development is a lifelong process. Yet the sources of these patterns are unknown. Theories suggest that personality continuity and change may result from environmentally mediated processes of identity development due to age-graded social roles and individual life experiences, but also from biological maturation. Genetically informative longitudinal studies across different age cohorts allow a differentiated picture of genetic and environmental sources. In this article, I give a short overview of the genetic and environmental contributions to rank-order continuity and change in neuroticism and extraversion. Both genetic and environmental factors contribute to personality continuity and change, but genetic factors affect rank-order change only in younger decades of life, whereas environmental influences appear to represent a lifelong source of interindividual differences in personality development.

Keywords

genetic continuity, environmental continuity, heritability, neuroticism, extraversion

> The science of behavioral genetics, largely through twin studies, has made a persuasive case that much of our identity is stamped on us from conception; to that extent, our lives seem to be pre-chosen—all we have to do is live out the script that is written in our genes . . . Still, it's confounding to imagine . . . any gardener knows that rather small variations of water and fertilizer and soil acidity affect the development of far less complex organisms—are humans so indifferent to the environment they are planted in?
>
> Wright, 1997, pp. 143-148

Personality traits are commonly defined as broad, dispositional, and comparative features of psychological individuality. The most prominent and extensively studied bipolar personality-trait continua are neuroticism–emotional stability and extraversion–introversion. Compared with more emotionally stable persons, individuals who are high on neuroticism are more likely to feel anxious, depressed, and guilty. They are more often self-conscious and are more likely to experience situations as threatening and stressful. Therefore, not surprisingly, neuroticism is linked to several psychopathological symptoms (Weinstock & Whisman, 2006). Individuals high on extraversion are more active, gregarious, and assertive than are introverts, who prefer to engage in fewer social activities and to maintain fewer social contacts. Extraverts also engage in more social and enterprising leisure-time and job-related interests (Kandler, Bleidorn, Riemann, Angleitner, & Spinath, 2011).

Neuroticism and extraversion, or roughly parallel trait conceptions, are often listed as the first two personality dimensions in models of personality or temperament (see Bouchard & Loehlin, 2001). Although facets of these traits, such as anxiety (a facet of neuroticism) and activity (a facet of extraversion), are already observable within the first years of life, a clear structure of personality traits does not begin to appear before the preteen period (McAdams & Olson, 2010).

Personality traits are enduring features accounting for consistencies in behavior, thoughts, and feelings across situations and occasions. Longitudinal studies of personality traits have provided robust evidence that the stability of interindividual

Corresponding Author:
Christian Kandler, Department of Psychology, Bielefeld University, Universitätsstraße 25, D-33615 Bielefeld, Germany
E-mail: christian.kandler@uni-bielefeld.de

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016

differences over time (referred to hereafter as *rank-order continuity*) increases with age. In other words, a person's specific position on a personality-dimension continuum relative to the positions of other people stabilizes with age. Stability coefficients (i.e., indices of rank-order continuity between occasions of personality assessment; these coefficients can vary from 0, *no continuity*, to 1, *perfect continuity*), which are low in childhood (approximately .30), rise across adolescence and young adulthood, reaching a plateau (approximately .70) between the ages of 40 and 70 (Roberts & DelVecchio, 2000; Terracciano, Costa, & McCrae, 2006). In old age, stability coefficients appear to decay (Lucas & Donnellan, 2011; Specht, Egloff, & Schmukle, 2011). Irrespective of the age group studied, stability coefficients decline as the time intervals between measurements of personality traits get longer (Fraley & Roberts, 2005). These patterns of stability and instability in individual ranks on personality dimensions point to the fact that personality development is a lifelong individual process. Although these patterns of rank-order continuity in personality are well known, their sources remain a matter of controversy.

McCrae and his colleagues (2000) formulated a theory of biological maturation (McCrae & Costa, 2008). According to this perspective, personality traits are stable, genetically (or at least biologically) anchored basic tendencies of thoughts, feelings, and behavior. Deviations from perfect continuity should be due to short-term environmentally induced fluctuations, but also to systematic and random measurement error. In fact, controlling for artifacts and occasion-specific effects results in higher heritability estimates (i.e., estimates of the degree to which relative differences among individuals are due to genetic factors) and larger stability coefficients (Kandler et al., 2010). Furthermore, similar mean-level trends (i.e., trait change over time averaged across individuals in a sample or population) for neuroticism and for extraversion have been found across different nations and cultures (McCrae et al., 1999, 2000). Neuroticism tends to decrease with age, whereas extraversion tends to be stable. Individual deviations from the mean-level trend (i.e., interindividual differences in intraindividual change) in neuroticism have been found to be genetically influenced (Bleidorn, Kandler, Riemann, Angleitner, & Spinath, 2009). These results support the hypothesis that genetically programmed maturation is a propulsive mechanism of personality development. Consequently, the increase in rank-order continuity of personality from childhood to adulthood may result from individual differences in biological maturation, and the decrease in such continuity in older age may be explained by individual differences in biological degeneration.

Other researchers have formulated different perspectives, attributing changes in personality to environmental sources, such as individual requirements in social roles (e.g., worker, spouse, parent) and related investments (Roberts & Wood, 2006), normative life transitions (e.g., departure from one's parental home, the completion of school or an apprenticeship, retirement), and individual life events that result in major changes of individuals' circumstances (e.g., accidents, the birth of one's own child, marriage). In other words, personality development and change are primarily attributable to age-graded social roles and related experiences (Roberts, Wood, & Caspi, 2008). Since the average direction of personality change (i.e., mean-level change in a population) is often positive with respect to social functionality (e.g., people become more emotionally stable with age), personality development is considered to be a process leading to social maturity (Roberts & Mroczek, 2008). From this perspective, the lower stability coefficients in younger ages compared with coefficients in middle adulthood are due to interindividual differences in the life transitions and role transfers that typically occur during the younger period of life, such as the completion of an apprenticeship and beginning of a job, departure from one's parental home, marriage, and new parenthood (Arnett, 2000). In fact, this period of life is strongly shaped by individually different developmental courses that are attributable to interindividual differences in level and duration of education, family formation, and economic security (McAdams & Olson, 2010). Social roles and contexts stabilize in midlife. They change again in old age until social adjustment breaks down in advanced old age. Consequently, the increase in rank-order continuity of personality traits from adolescence to adulthood, the plateau in middle adulthood, and the decrease of continuity in older age may result from interindividual differences in stabilizing, maintaining, and destabilizing one's personal identity in life.

On the basis of these theoretical considerations, I suggest an integrative perspective. That is, both genetic and environmental influences may explain personality development, at least in the case of the rank-order continuity of personality traits, and at least in the case of neuroticism and extraversion. Longitudinal behavioral genetics studies (e.g., longitudinal twin or adoption studies) on different age cohorts have allowed for deeper insight. In the following sections, I briefly review and discuss the results from these studies with respect to the patterns of genetic and environmental effects on rank-order continuity in neuroticism and extraversion.

The Role of Genetic Factors in Personality Continuity and Change

Using genetically informative designs, the interindividual differences in trait scores that are due to genetic differences (i.e., heritability) and to environmental influences can be disentangled. Genetically informative longitudinal designs allow a disentanglement of genetic and environmental components on each of the measurement occasions, as well as the calculation of the level of continuity of each of these components (i.e., genetic and environmental continuity).

It has been consistently shown that genetic continuity is far from perfect (i.e., stability coefficients are consistently less than 1) during childhood and adolescence (De Fruyt et al., 2006; Gillespie, Evans, Wright, & Martin, 2004; Spengler,

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016

Gottschling, & Spinath, 2012). That is, across childhood and adolescence, new genetic factors emerge that contribute to interindividual differences and rank-order changes in personality. Like rank-order continuity, genetic continuity tends to increase across adolescence and young adulthood (Hopwood et al., 2011; Viken, Rose, Kaprio, & Koskenvuo, 1994) until it reaches almost perfect stability in middle and late adulthood (W. Johnson, McGue, & Krueger, 2005; Pedersen, 1993; Read, Vogler, Pedersen, & Johansson, 2006).

This pattern of genetic continuity has been found for neuroticism and extraversion or related variables (see green lines in Fig. 1 for neuroticism and in Fig. 2 for extraversion). Therefore, the steady increase in observable rank-order continuity from childhood to adulthood can be accounted for by stabilizing interindividual genetic differences. The relatively high stability of interindividual personality differences in middle adulthood is accompanied by stability in genetic differences. In other words, rank-order continuity in neuroticism and extraversion can be explained by interindividual differences in genetically influenced maturation.

The Role of Environmental Factors in Personality Continuity and Change

Stable genetic differences among individuals after age 40, however, cannot account for the decay of stability coefficients in older age. Therefore, nongenetic factors may also contribute to rank-order continuity. Across a broad spectrum of genetically informative studies, A. M. Johnson, Vernon, and Feiler (2008) found that the primary nongenetic sources of interindividual differences in neuroticism and extraversion are specific to individuals (i.e., not shared by family members reared together). These effects are known as *nonshared environmental effects*. From a behavioral genetics perspective, "environmental" effects include physiologically and biochemically mediated effects, such as environmentally activated epigenetic influences. Nonshared environmental factors show different degrees of stability (i.e., environmental continuity) with respect to the age studied.

Studying twins between the ages of 12 and 16, Gillespie et al. (2004) found rather low coefficients for environmental continuity for both extraversion (.17 for females and .18 for males) and neuroticism (.12 for males and .36 for females). Hopwood and his colleagues (2011) studied the transition from adolescence to adulthood (age 17 to 24). The researchers found environmental-continuity coefficients of .36 for negative emotionality (neuroticism), .37 for communal positive emotionality, and .39 for agentic positive emotionality (the latter two traits can be seen as facets of extraversion). They also investigated rank-order continuity between the ages of 24 and 29. For this interval, they found higher environmental continuity, with coefficients ranging from .56 to .60. For

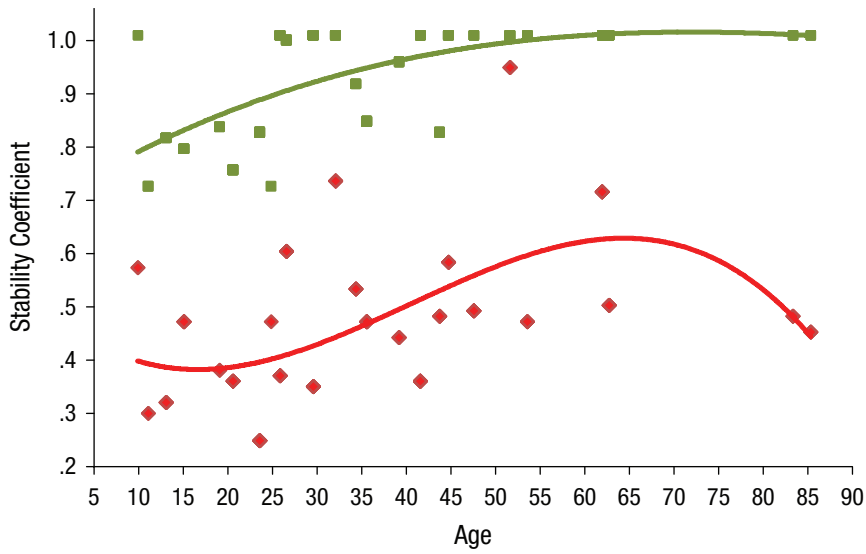

Fig. 1. Genetic continuity (depicted by the green line and data points) and environmental continuity (depicted by the red line and data points) of interindividual differences in neuroticism (or negative emotionality) as a function of age. Higher stability coefficients (i.e., indices of continuity between assessments) indicate greater stability over time. The stability coefficients shown are based on the results of genetically informative longitudinal studies (weighted by sample sizes): Bratko and Butkovic (2007), De Fruyt et al. (2006), Gillespie, Evans, Wright, and Martin (2004), Hopwood et al. (2011), W. Johnson, McGue, and Krueger (2005), Kandler et al. (2010), McGue, Bacon, and Lykken (1993), Pedersen (1993), Read, Vogler, Pedersen, and Johansson (2006), Spengler, Gottschling, and Spinath (2012), Viken, Rose, Kaprio, and Koskenvuo (1994), and Wray, Birley, Sullivan, Visscher, and Martin (2007). Across studies and sample cohorts, intervals between assessments ranged between 2 and 10 years (M = 5.15 years).

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016

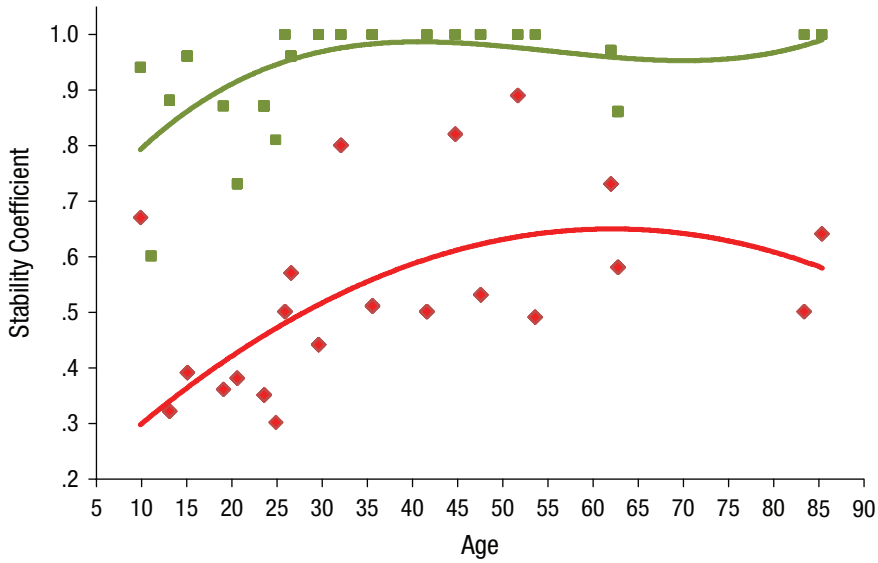

Fig. 2. Genetic continuity (depicted by the green line and data points) and environmental continuity (depicted by the red line and data points) of interindividual differences in extraversion (or positive emotionality) as a function of age. Higher stability coefficients (i.e., indices of continuity between assessments) indicate greater stability over time. The stability coefficients shown are based on the results of genetically informative longitudinal studies (weighted by sample sizes): Bratko and Butkovic (2007), De Fruyt et al. (2006), Gillespie, Evans, Wright, and Martin (2004), Hopwood et al. (2011), W. Johnson, McGue, and Krueger (2005), Kandler et al. (2010), McGue, Bacon, and Lykken (1993), Pedersen (1993), Read, Vogler, Pedersen, and Johansson (2006), Spengler, Gottschling, and Spinath (2012), and Viken, Rose, Kaprio, and Koskenvuo (1994). Across studies and sample cohorts, intervals between assessments ranged between 2 and 10 years (M = 4.88 years).

middle adulthood, W. Johnson and colleagues (2005) reported 5-year environmental-continuity coefficients over .70. However, other studies have indicated that environmental continuity decreases in later adulthood (Pedersen & Reynolds, 1998; Read et al., 2006).

These findings and results from other genetically informative studies (see red lines in Figs. 1 and 2) point to the fact that changing levels of rank-order continuity in neuroticism and extraversion are also attributable to changing levels of environmental continuity across the life span. Moreover, the decrease in rank-order continuity in older age appears to be exclusively due to a decline in the stability of environmental factors. Therefore, interindividual differences in experiences of major life transitions or major life events (whether temporal, quantitative, or qualitative) may contribute to personality change and, thus, to personality development. Environmental contributions to change in interindividual ranks may occur across the entire life span, but they are more pronounced in younger ages (because of events such as leaving one's parental home, starting a job, and forming a new family) and older ages (because of events such as retirement, illness, death of spouses or friends). Because "environmental" factors also include biologically mediated effects, the declining rank-order continuity, especially in later adulthood, might be due to biological causes (e.g., multi-infarct dementia).

The Interplay of Genetic and Environmental Factors

As life narratives and memories cumulate across the life span, effects of life experiences may cumulate, too. Thus, environmental effects on interindividual differences in neuroticism and extraversion may cumulate with age, leading to an increase in environmental variance (McCartney, Harris, & Bernieri, 1990). If interindividual differences due to environmental effects increase with age, stable interindividual differences accounted for by genetic effects should proportionally decrease. Consequently, heritability estimates for personality traits should decrease with age.

Most genetically informative studies that have assessed heritability estimates for different age cohorts (e.g., Loehlin & Martin, 2001) have not found significant differences between age groups. However, other studies that have combined cross-sectional data (from different age cohorts) and longitudinal data have suggested that heritability estimates decrease with age (Kandler et al., 2010; Viken et al., 1994). A meta-analytic review of fifteen genetically informative studies on different age groups showed that the level of heritability of neuroticism and extraversion appears to depend on the age studied (see Fig. 3). Heritability of extraversion tends to increase until people are in their early 40s, but then it declines continuously. For

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016

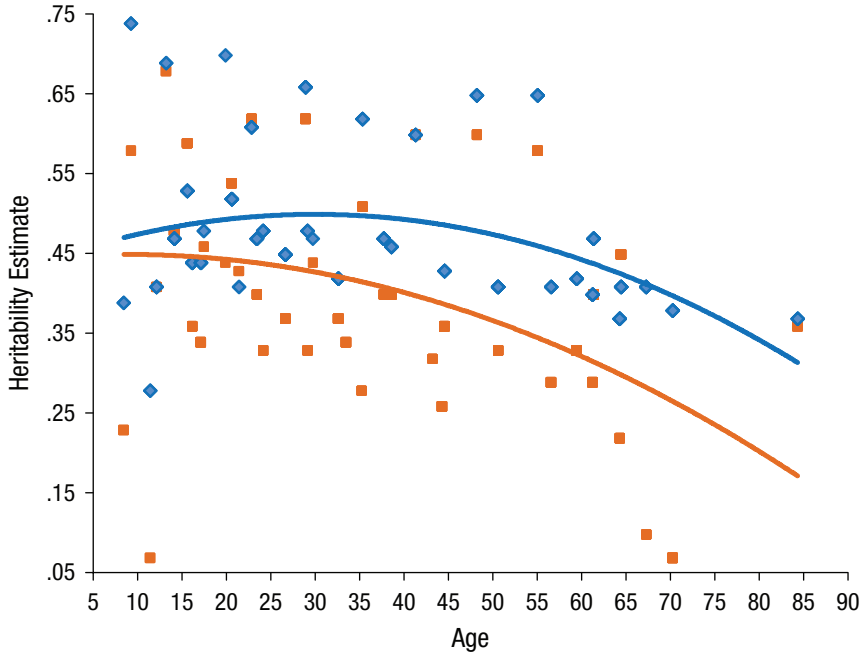

Fig. 3. Heritability estimates (the proportion of interindividual differences accounted for by genetic differences) for extraversion (depicted by the blue line and data points) and neuroticism (depicted by the orange line and data points) as a function of age. The heritability estimates shown are based on the results of genetically informative longitudinal and age-cohort studies (weighted by sample size): Bratko and Butkovic (2007), De Fruyt et al. (2006), Gillespie, Evans, Wright, and Martin (2004), Hopwood et al. (2011), W. Johnson, McGue, and Krueger (2005), Kandler et al. (2010), Loehlin and Martin (2001), McGue, Bacon, and Lykken (1993), Pedersen and Reynolds (1998), Read, Vogler, Pedersen, and Johansson (2006), Rettew et al. (2006), Rettew, Rebollo-Mesa, Hudziak, Willemsen, and Boomsma (2008), Spengler, Gottschling, and Spinath (2012), Viken, Rose, Kaprio, and Koskenvuo (1994), and Wray et al. (2007).

neuroticism, heritability progressively decreases after people enter adulthood.

The changing importance of genetic contributions to interindividual differences in neuroticism and extraversion with age differs from the pattern observed with regard to cognitive abilities. For cognitive abilities, heritability continuously increases with age (W. Johnson, 2010). One reason for this distinction may be that genes and environmental effects differentially contribute to how a person typically acts (i.e., his or her temperament or narrow-sense personality) and how capable a person is (i.e., his or her abilities or skills).

An increase in the level of personality traits' heritability may reflect the increasing importance of active and reactive gene-environment correlations. People actively select, create, and change their environments (e.g., by changing their friends, residences, or jobs) or evoke social reactions that match their genetically predisposed traits. Environments and social reactions, in turn, may affect personality traits. For example, individuals who are genetically predisposed to be more extraverted may perceive life events as more controllable and positive; experiencing life events as controllable and positive, in turn, may augment such people's levels of extraversion (Kandler, Bleidorn, Riemann, Angleitner, & Spinath, 2012). That is, genetic influences may partly drive the environmental effects accounting for the increasing heritability. The processes involved in these gene-environment correlations may be propulsive mechanisms of the development of cognitive abilities across the life span and the development of extraversion in the first decades of life.

The increasing heritability of extraversion until age 30 and the cumulatively decreasing heritability of neuroticism and extraversion after age 30 may accompany midlife tipping points. For example, young adults often try to expand the self and gain new information—they actively try to change their environments to match their goal pursuits; by contrast, older adults more often try to stabilize and protect the quality of their lives and ongoing relationships—they change the self to fit the environment (see McAdams & Olson, 2010, for a review). Consequently, genetic and environmental effects do not act separately; rather, they interact in complex ways that account for the development of personality traits, and the contribution of life experiences to personality development cumulates with age.

Conclusion

I hope that this short review will open the door for further research identifying the mechanisms of the gene-environment

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016

interplay in personality development and also focusing on the patterns and sources of other personality traits (e.g., openness, agreeableness, and conscientiousness). More genetically and environmentally informative research is sorely needed. At this point, we know that genetic factors stabilize with age, leading to an increase in rank-order continuity of neuroticism and extraversion and to personality formation across childhood, adolescence, and young adulthood. Environmental sources contribute to this increasing continuity but also to decreasing continuity in old age. Moreover, environmental influences appear to cumulate across the life span, leading to a decline of the genetic contribution to variance in neuroticism and extraversion with age, a pattern pointing to lifelong changes in personality due to environmental influences. We develop into the people we are predisposed to become because of our genetic makeup, but to a large degree, the development of or changes in our personality will be formed by the environment.

Recommended Reading

Bouchard, T. J., & Loehlin, J. C. (2001). (See References). A comprehensive review of research on personality traits and related variables from a behavioral genetics perspective.

Fraley, C., & Roberts, B.W. (2005). (See References). A comprehensive work on patterns of continuity described as mathematical models.

McAdams, D. P., & Olson, B. D. (2010). (See References). A clearly written, user-friendly, and empirically founded theoretical perspective on personality development of different levels of personality elements, such as traits, motives, and narratives.

McCrae, R. R., & Costa, P. T. (2008). (See References). A comprehensive overview of research and theory on personality traits as biologically anchored basic tendencies.

Wright, L. (1997). (See References). An interesting and reader-friendly book on the history of twin studies and the secrets of our identity.

Declaration of Conflicting Interests

The author declared that he had no conflicts of interest with respect to his authorship or the publication of this article.

References

Arnett, J. J. (2000). Emerging adulthood: A theory of development from the late teens through the twenties. *American Psychologist, 55*, 469–480.

Bleidorn, W., Kandler, C., Riemann, R., Angleitner, A., & Spinath, F. (2009). Patterns and sources of adult personality development: Growth curve analyses of the NEO-PI-R scales in a longitudinal twin study. *Journal of Personality and Social Psychology, 97*, 142–155.

Bouchard, T. J., & Loehlin, J. C. (2001). Genes, evolution, and personality. *Behavior Genetics, 31*, 243–273.

Bratko, D., & Butkovic, A. (2007). Stability and genetic and environmental effects from adolescence to young adulthood: Results of Croatian longitudinal twin study of personality. *Twin Research and Human Genetics, 10*, 151–157.

De Fruyt, F., Bartels, M., Van Leeuwen, K. G., De Clercq, B., Decuyper, M., & Mervielde, I. (2006). Five types of personality continuity in childhood and adolescence. *Journal of Personality and Social Psychology, 91*, 538–552.

Fraley, C., & Roberts, B. W. (2005). Patterns of continuity: A dynamic model for conceptualizing the stability of individual differences in psychological constructs across the life course. *Psychological Review, 112*, 60–74.

Gillespie, N. A., Evans, D. E., Wright, M. M., & Martin, N. G. (2004). Genetic simplex modeling of Eysenck's dimensions of personality in a sample of young Australian twins. *Twin Research, 7*, 637–648.

Hopwood, C. J., Donnellan, M. B., Krueger, R. F., McGue, M., Iacono, W. G., Blonigen, D. M., & Burt, S. A. (2011). Genetic and environmental influences on personality trait stability and growth during the transition to adulthood: A three-wave longitudinal study. *Journal of Personality and Social Psychology, 100*, 545–556.

Johnson, A. M., Vernon, P. A., & Feiler, A. R. (2008). Behavioral genetic studies of personality: An introduction and review of the results of 50+ years of research. In G. J. Boyle, G. Matthews, & D. H. Saklofske (Eds.), *The SAGE handbook of personality theory and assessment: Personality theories and models* (Vol. 1, pp. 145–173). London, England: Sage.

Johnson, W. (2010). Understanding the genetics of intelligence: Can height help? Can corn oil? *Current Directions in Psychological Science, 19*, 177–182.

Johnson, W., McGue, M., & Krueger, R. F. (2005). Personality stability in late adulthood: A behavioral genetic analysis. *Journal of Personality, 73*, 523–551.

Kandler, C., Bleidorn, W., Riemann, R., Angleitner, A., & Spinath, F. M. (2011). The genetic links between the big five personality traits and general interest domains. *Personality and Social Psychology Bulletin, 37*, 1633–1643.

Kandler, C., Bleidorn, W., Riemann, R., Angleitner, A., & Spinath, F. M. (2012). Life events as environmental states and genetic traits and the role of personality: A longitudinal twin study. *Behavior Genetics, 42*, 57–72.

Kandler, C., Bleidorn, W., Riemann, R., Spinath, F. M., Thiel, W., & Angleitner, A. (2010). Sources of cumulative continuity in personality: A longitudinal multiple-rater twin study. *Journal of Personality and Social Psychology, 98*, 995–1008.

Loehlin, J. C., & Martin, N. G. (2001). Age changes in personality traits and their heritabilities during the adult years: Evidence from Australian Twin Registry samples. *Personality and Individual Differences, 30*, 1147–1160.

Lucas, R. E., & Donnellan, M. B. (2011). Personality development across the life span: Longitudinal analyses with a national sample from Germany. *Journal of Personality and Social Psychology, 101*, 847–861.

McAdams, D. P., & Olson, B. D. (2010). Personality development: Continuity and change over the life course. *Annual Review of Psychology, 61*, 5.1–5.26.

McCartney, K., Harris, M. J., & Bernieri, F. (1990). Growing up and growing apart: A developmental meta-analysis of twin studies. *Psychological Bulletin, 107*, 226–237.

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016

McCrae, R. R., & Costa, P. T., Jr. (2008). The five-factor theory of personality. In O. P. John, R. W. Robins, & L. A. Pervin (Eds.), *Handbook of personality: Theory and research* (3rd ed., pp. 159–181). New York, NY: Guilford.

McCrae, R. R., Costa, P. T., Jr., Ostendorf, F., Angleitner, A., Hrebrícková, M., Avia, M. D., . . . Smith, B. P. (2000). Nature over nurture: Temperament, personality, and life span development. *Journal of Personality and Social Psychology, 78*, 173–186.

McCrae, R. R., Costa, P. T., Jr., Pedroso de Lima, M., Simões, A., Ostendorf, F., Angleitner, A., . . . Piedmont, R. L. (1999). Age differences in personality across the adult life span: Parallels in five cultures. *Developmental Psychology, 35*, 466–477.

McGue, M., Bacon, S., & Lykken, D. T. (1993). Personality stability and change in early adulthood: A behavioral genetic analysis. *Developmental Psychology, 29*, 96–109.

Pedersen, N. L. (1993). Genetic and environmental continuity and change in personality. In T. J. Bouchard Jr. & P. Propping (Eds.), *Twins as a tool of behavioral genetics* (pp. 147–162). New York, NY: Wiley.

Pedersen, N. L., & Reynolds, C. A. (1998). Stability and change in adult personality: Genetic and environmental components. *European Journal of Personality, 12*, 365–386.

Read, S., Vogler, G. P., Pedersen, N. L., & Johansson, B. (2006). Stability and change in genetic and environmental components of personality in old age. *Personality and Individual Differences, 40*, 1637–1647.

Rettew, D. C., Rebollo-Mesa, I., Hudziak, J. J., Willemsen, G., & Boomsma, D. I. (2008). Non-additive and additive genetic effects on extraversion in 3314 Dutch adolescent twins and their parents. *Behavior Genetics, 38*, 223–233.

Rettew, D. C., Vink, J. M., Willemsen, G., Doyle, A., Hudziak, J. J., & Boomsma, D. I. (2006). The genetic architecture of neuroticism in 3301 Dutch adolescent twins as a function of age and sex: A study from the Dutch twin register. *Twin Research and Human Genetics, 9*, 24–29.

Roberts, B. W., & DelVecchio, W. F. (2000). The rank-order consistency of personality from childhood to old age: A quantitative review of longitudinal studies. *Psychological Bulletin, 126*, 3–25.

Roberts, B. W., & Mroczek, D. (2008). Personality trait change in adulthood. *Current Directions in Psychological Science, 17*, 31–35.

Roberts, B. W., & Wood, D. (2006). Personality development in the context of the Neo-Socioanalytic Model of personality. In D. Mroczek & T. Little (Eds.), *Handbook of personality development* (pp. 11–39). Mahwah, NJ: Erlbaum.

Roberts, B. W., Wood, D., & Caspi, A. (2008). The development of personality traits in adulthood. In O. P. John, R. W. Robins, & L. A. Pervin (Eds.), *Handbook of personality: Theory and research* (3rd ed., pp. 375–398). New York, NY: Guilford.

Specht, J., Egloff, B., & Schmukle, S. C. (2011). Stability and change of personality across the life course: The impact of age and major life events on mean-level and rank-order stability of the big five. *Journal of Personality and Social Psychology, 101*, 862–882.

Spengler, M., Gottschling, J., & Spinath, F. M. (2012). Personality in childhood: A longitudinal behavior genetic approach. *Personality and Individual Differences, 53*, 411–416.

Terracciano, A., Costa, P. T., Jr., & McCrae, R. R. (2006). Personality plasticity after age 30. *Personality and Social Psychology Bulletin, 32*, 999–1009.

Viken, R. J., Rose, R. J., Kaprio, J., & Koskenvuo, M. (1994). A developmental genetic analysis of adult personality: Extraversion and neuroticism from 18 to 59 years of age. *Journal of Personality and Social Psychology, 66*, 722–730.

Weinstock, L. M., & Whisman, M. A. (2006). Neuroticism as a common feature of the depressive and anxiety disorders: A test of the revised integrative hierarchical model in a national sample. *Journal of Abnormal Psychology, 115*, 68–74.

Wray, N. R., Birley, A. J., Sullivan, P. F., Visscher, P. M., & Martin, N. G. (2007). Genetic and phenotypic stability of measures of neuroticism over 22 years. *Twin Research and Human Genetics, 10*, 695–702.

Wright, L. (1997). *Twins: And what they tell us about who we are*. New York, NY: Wiley.

Downloaded from cdp.sagepub.com at PENNSYLVANIA STATE UNIV on September 19, 2016